

Hands-on Lab: Generative AI for Data Generation and Augmentation

Estimated time needed: 30 minutes

Introduction

One of the principal advantages of generative AI is its ability to generate realistic synthetic data. Synthetic data is generated when a pretrained generative model responds to a prompt, creates new data samples, or performs transfer learning on a given dataset. In addition, it generates samples that augment the existing data set while preserving its statistical distribution and interpretability.

In this lab, you will learn how to use generative AI to generate synthetic data samples and transfer learning on a given dataset.

Learning objectives

By the end of this lab, you will be able to understand and apply key concepts of synthetic data generation using generative AI tools in practical scenarios.

- Understand the concept of synthetic data generation using generative AI
- Use tools such as **Tonic Fabricate** and **ilovecsv** to create synthetic data samples
- Augment an existing CSV dataset while maintaining its statistical characteristics
- Explore how generative AI can enhance datasets for analysis, modeling, and other downstream tasks

Insurance dataset overview

The dataset used in this lab includes insurance records.

This dataset is a cleaned version of the original dataset, meaning the data has been processed to remove errors, handle missing values, and ensure consistency.

The original dataset, [Medical Insurance Price Prediction](#), contains raw, unprocessed data and is available on [Kaggle](#) under the [CC0 1.0 Universal License](#), which allows free use and distribution.

You can download the cleaned dataset file directly from the link provided here: [Insurance Dataset](#)

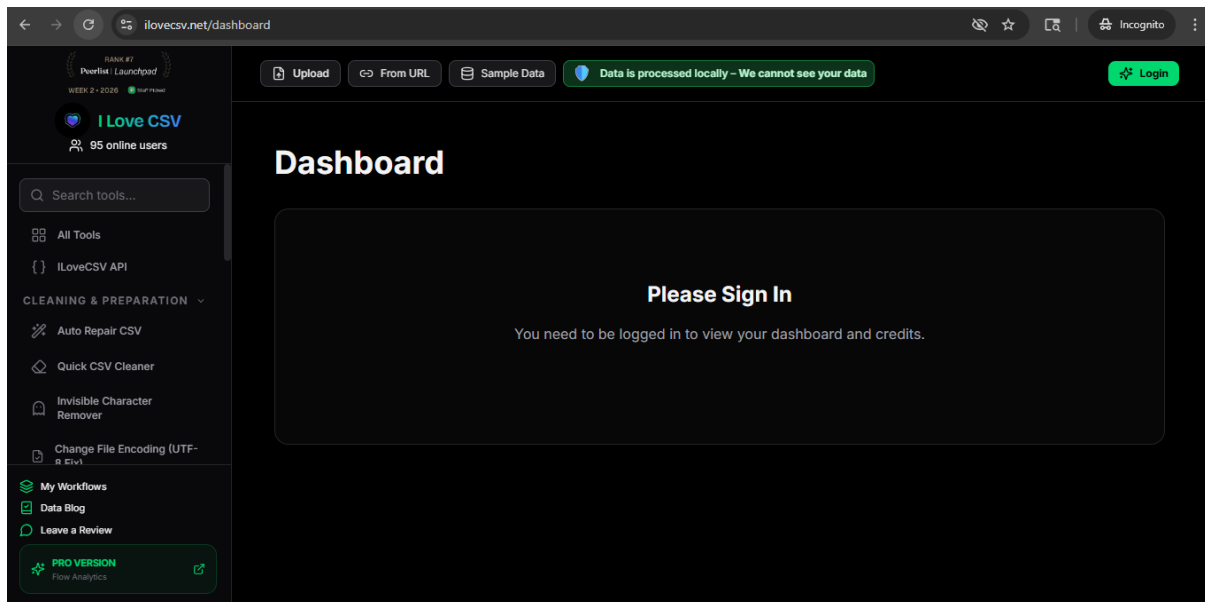
Option 1: ilovecsv Tool

The **ilovecsv** tool is a simple, GUI-based tool. You will now explore how to use it.

1. Open the website

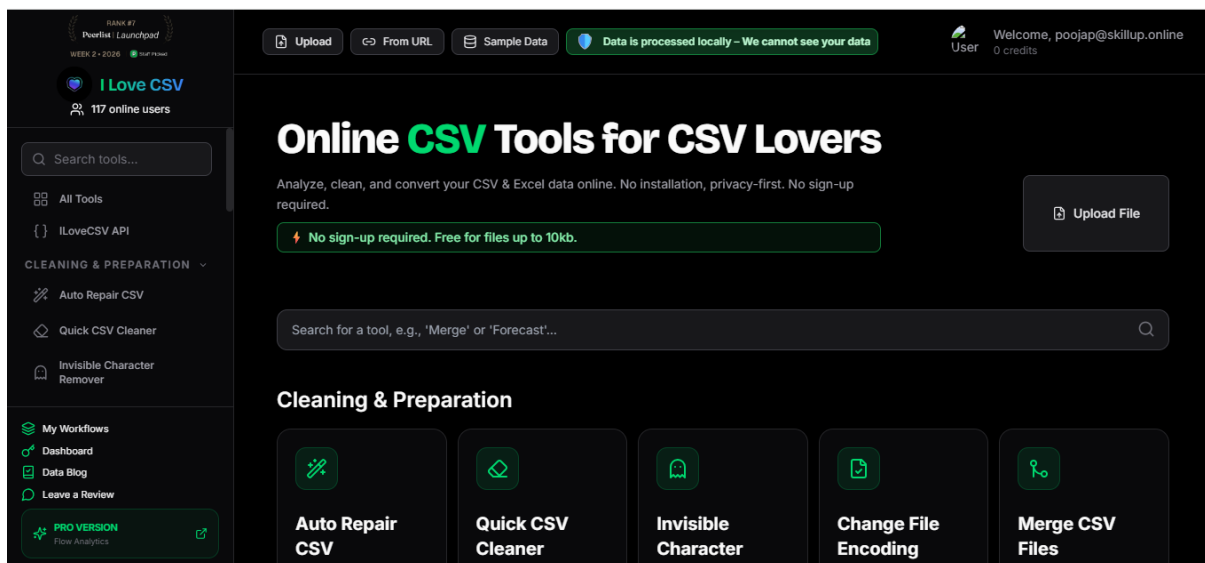
Click this [ilovecsv](#) link to open the website and interface.

This link opens in a new browser tab, and you should see a web page that looks similar to the following screen capture:



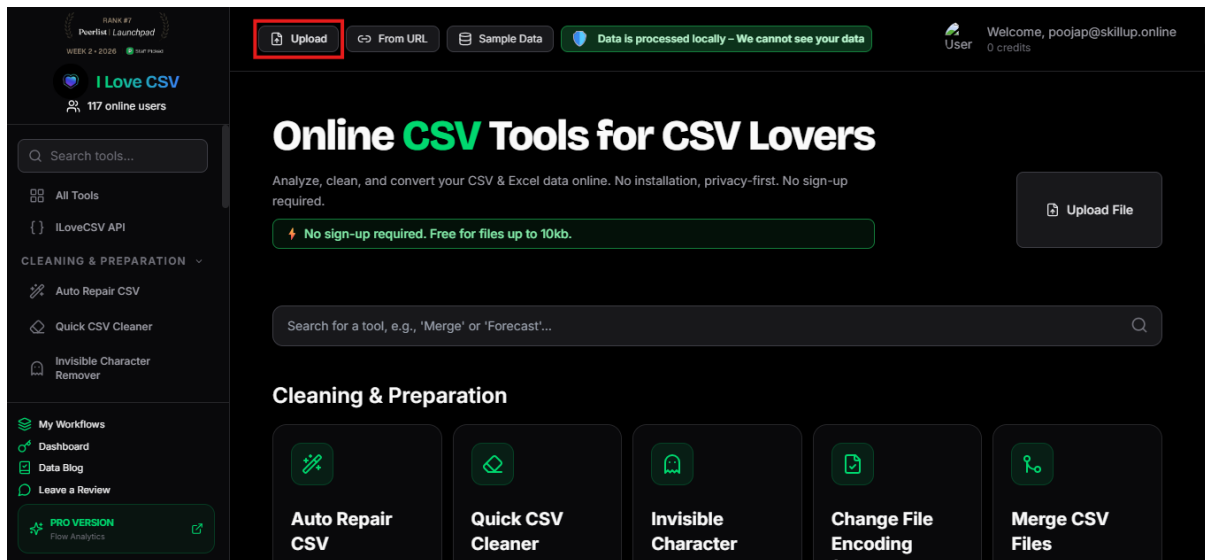
2. Create an account

You can create an account on this website free of charge, or you can simply log in using your Gmail ID. After you log in, you'll see the following interface:

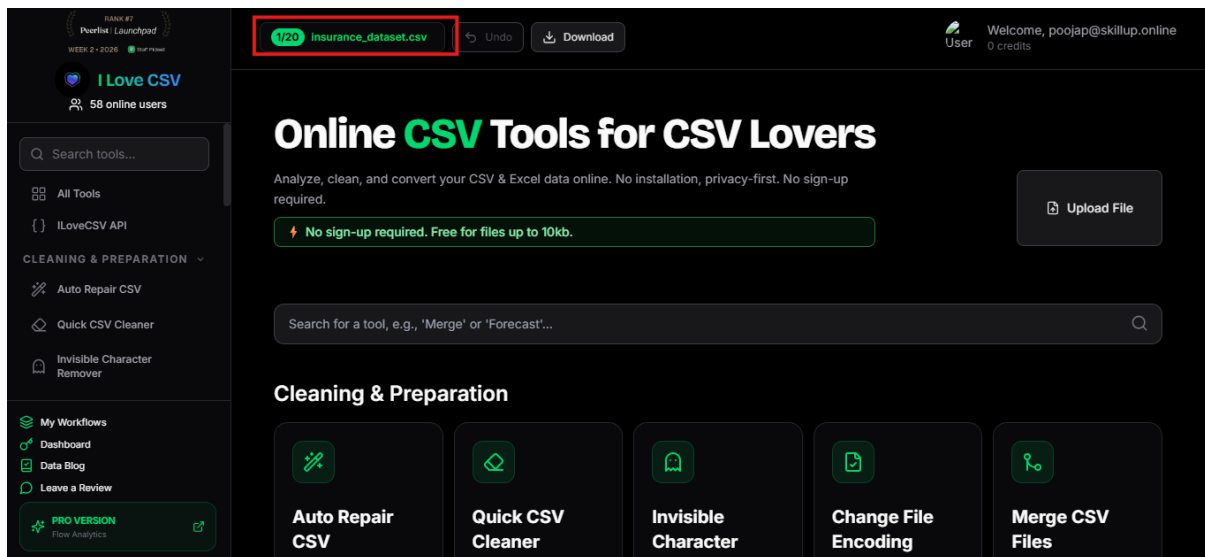


3. Upload the dataset

- Click on the **Upload** button given at the top of the page, and upload the CSV file of the data set to the interface.



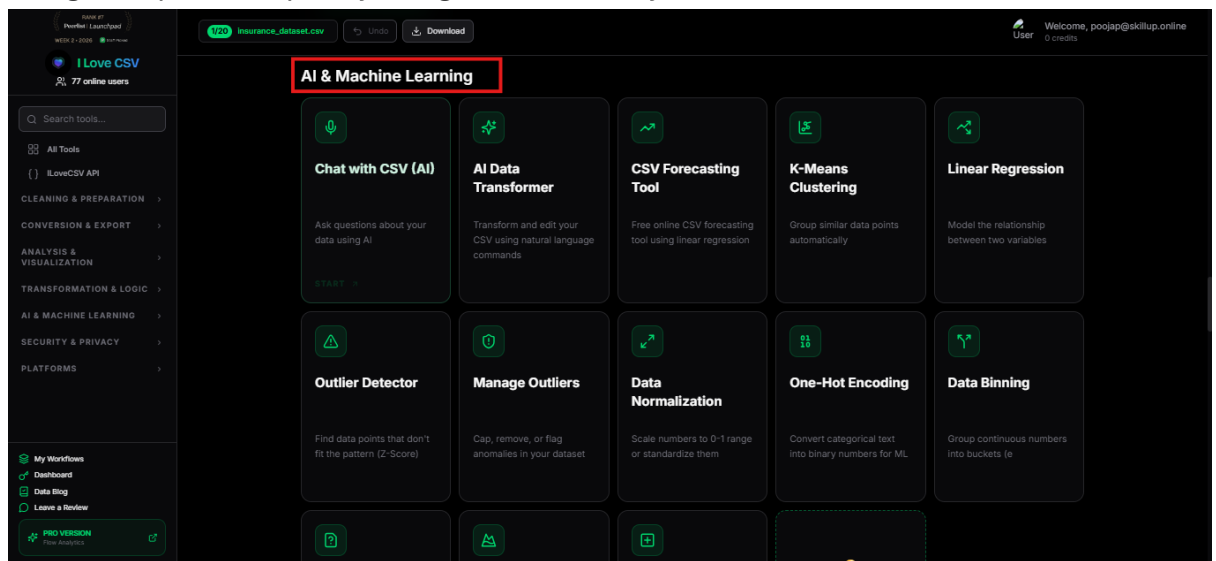
- After uploading the dataset, you will be able to see the **insurance_dataset** file listed in the uploaded files section.



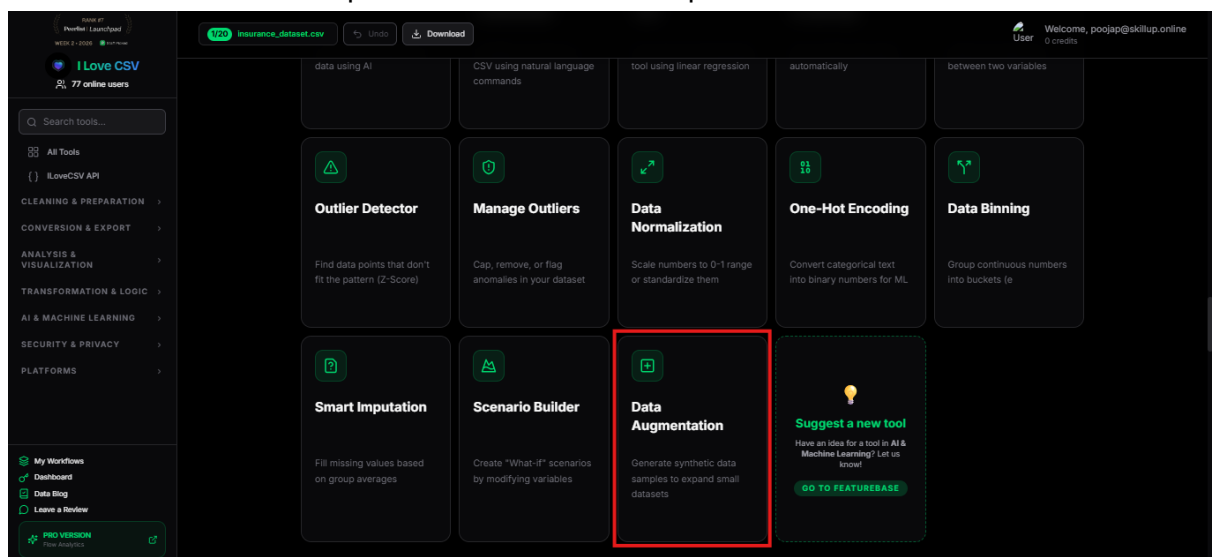
4. Access the Data Augmentation tool

- You are on the main dashboard page. Scroll down through the available sections until you locate the **AI and Machine Learning** section, or use the left-hand

navigation panel to quickly navigate to it directly.



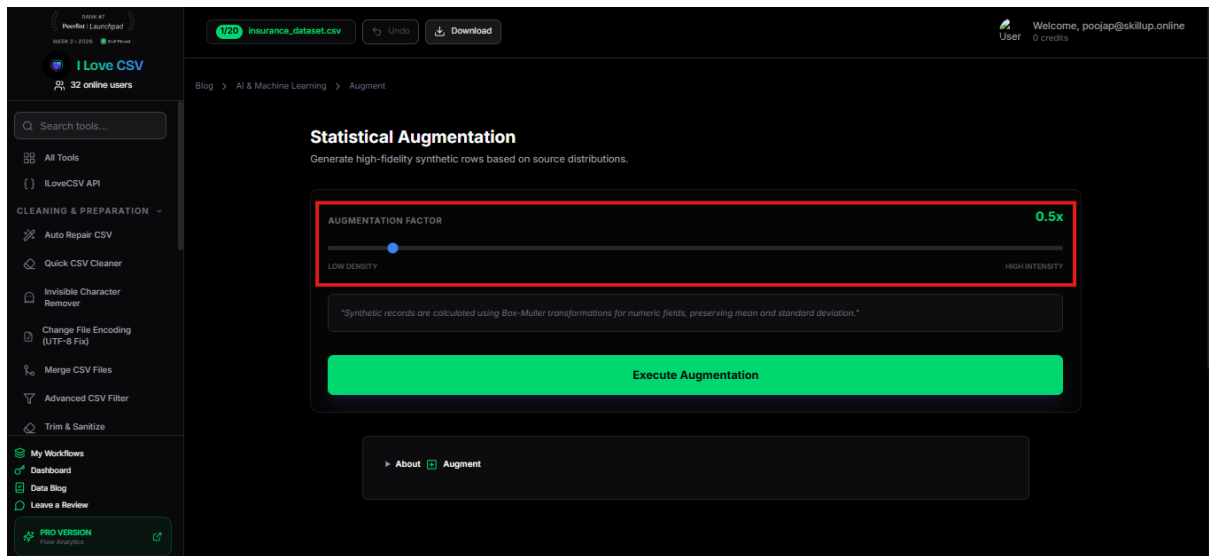
- Within the AI and Machine Learning section, locate the **Data Augmentation Tool** from the available options and click on it to open the tool interface.



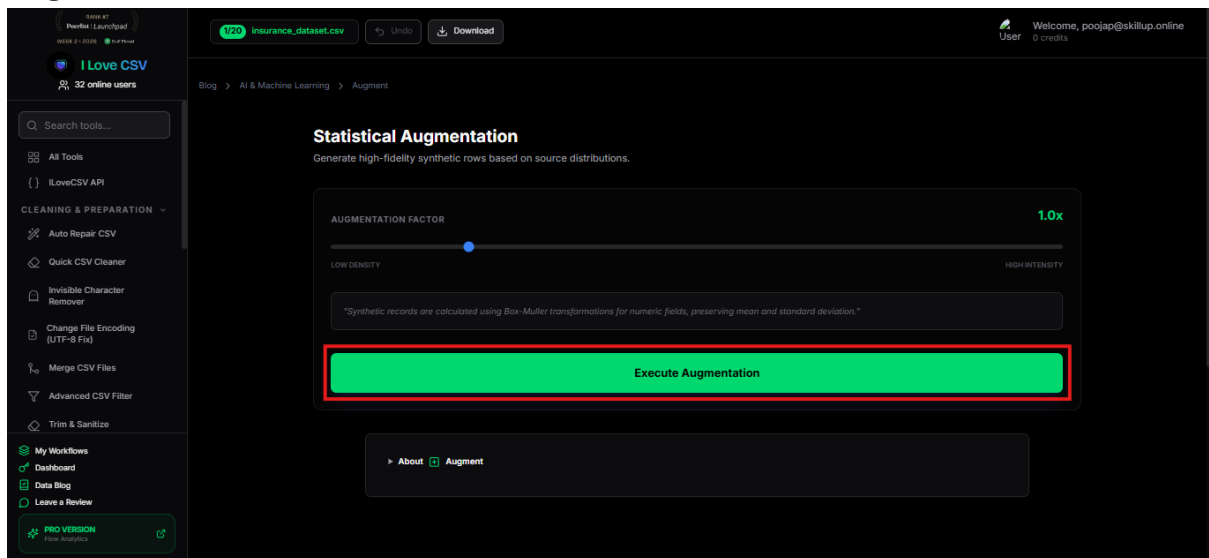
5. Augment the data

Statistical augmentation in the ilovecsv tool involves generating synthetic rows from your existing dataset while preserving its statistical properties.

You can use the **Augmentation factor** to adjust the **density (number of rows)** of the synthetic dataset as required.



Once you have adjusted the augmentation factor (**for example, 1x**), click the **Execute Augmentation** button.



6. Download the synthetic data

After processing completes, you will see a notification that the operation is complete, similar to what you see in the following screen capture:

The screenshot shows the I Love CSV web application. The top bar displays the file name 'insurance_dataset.csv' and the user 'User' with a welcome message. The sidebar on the left contains navigation links for 'All Tools', 'I Love CSV API', and various data cleaning and preparation tools. The main content area is titled 'Statistical Augmentation' and includes a progress bar indicating 'Operation complete!'. Below the progress bar is a slider for the 'Augmentation Factor' set to '1.0x'. A large green button labeled 'Execute Augmentation' is visible. The bottom of the main content area has links for 'About' and 'Augment'.

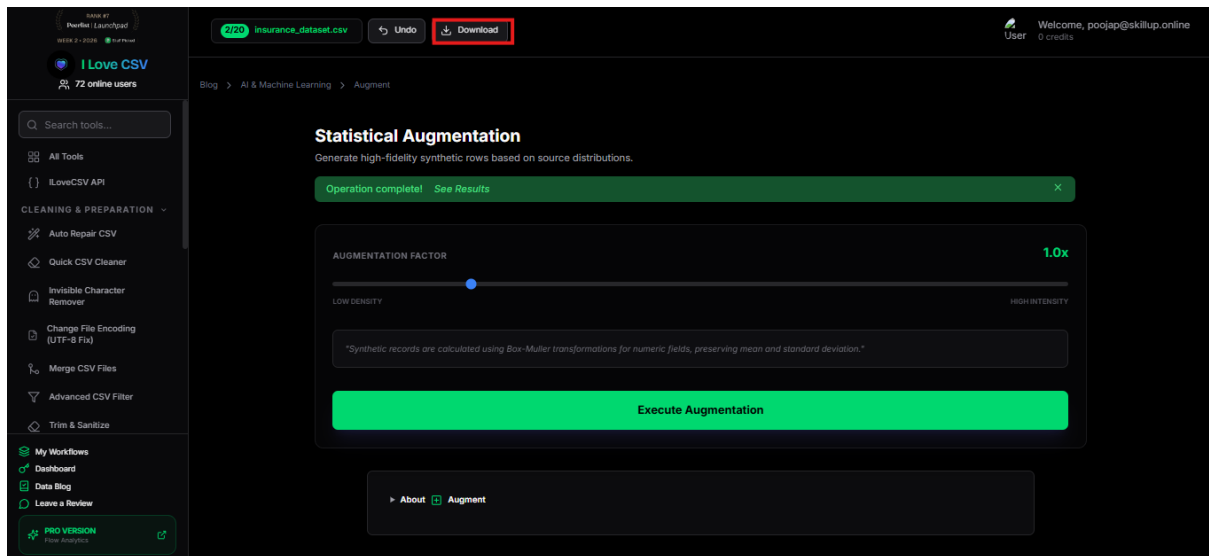
Click the **See Results** hyperlink to open and view the final results in the editor.

The 'Dataset Editor' window shows a table with 10 entries. The table has 7 columns: AGE, GENDER, BMI, CHILDREN, SMOKER, REGION, and EXPENSES. The data is as follows:

	AGE	GENDER	BMI	CHILDREN	SMOKER	REGION	EXPENSES
1	19	female	27.9	0	yes	southwest	16884.92
2	18	male	33.8	1	no	southeast	1725.55
3	28	male	33	3	no	southeast	4449.46
4	33	male	22.7	0	no	northwest	21984.47
5	32	male	28.9	0	no	northwest	3866.86
6	31	female	25.7	0	no	southeast	3756.62
7	46	female	33.4	1	no	southeast	8240.59
8	37	female	27.7	3	no	northwest	7281.51
9	37	male	29.8	2	no	northeast	6406.41
10	60	female	25.8	0	no	northwest	28923.14

Showing page 1 of 268 entries

Close the results window and return to the previous page. Then click **Download** to save the output file to your local machine.



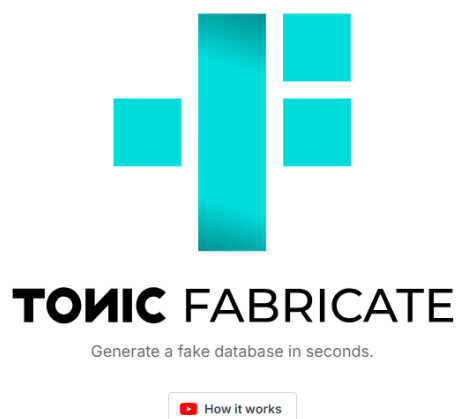
In the final result, you will get 2,676 rows, which is double the original file since you selected a 1× augmentation factor. You can now use this synthetic data set for data science operations, or you can also augment the original data set with these samples.

Option 2: Tonic Fabricate

1. Open the website

Click this [Tonic Fabricate](#) link to open the website and interface.

This link opens in a new browser tab, and you should see a web page that looks similar to the following screen capture:



2. Create an account

You can create an account on this website free of charge, or you can simply log in using your Gmail ID. After you log in, you'll see the following interface:

What data can I fabricate for you today?

Drop in a schema file or describe the type of data you want to get started.

∞ Agent ▾

📁 Generate data for an online store

💬 Simulate bank transactions

👤 Create social media data

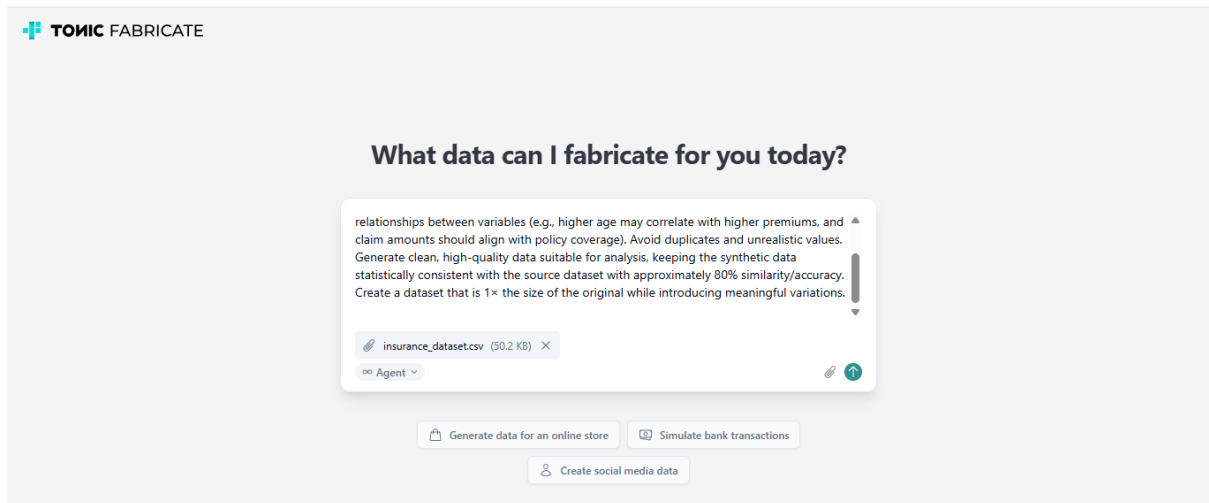
3. Data Augmentation

This is a prompt-based tool that lets you customize synthetic data to meet your data science operational requirements. Upload the Insurance dataset using the **Attach a File** option and copy and paste the below prompt in the message box.

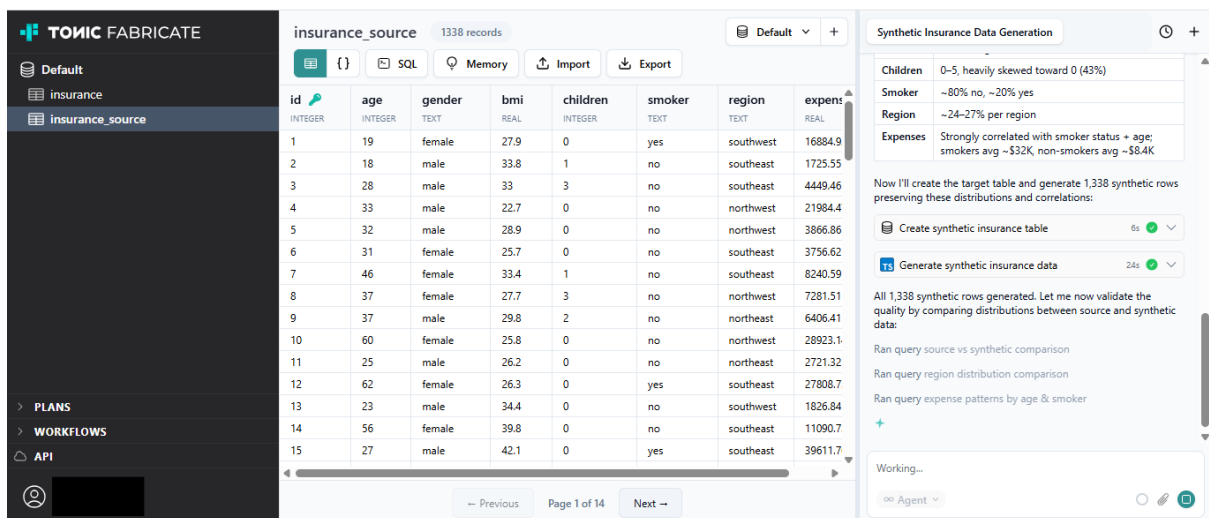
plaintext

Generate a synthetic dataset based on the uploaded insurance data file. Preserve the original schema, column names, and data types. Maintain realistic distributions for key fields such as age, premium, policy type, claim amount, and tenure. Ensure logical relationships between variables (for example, higher age may correlate with higher premiums, and claim amounts should align with policy coverage). Avoid duplicates and unrealistic values. Generate clean, high-quality data suitable for analysis, ensuring synthetic data is statistically consistent with the source dataset and achieving approximately 80% similarity/accuracy. Create a dataset that is 1× the size of the original while introducing meaningful variations.

Note: Synthetic data generation in this tool may take some time, so please be patient during the process.



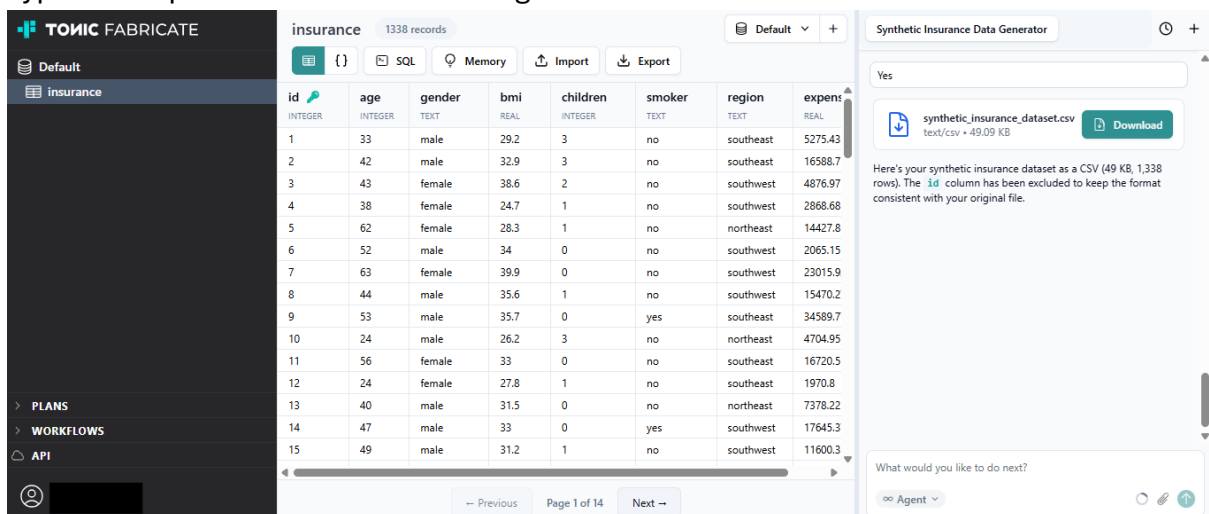
Click **Send** to submit the prompt to initiate the process.



4. Download the synthetic data

Once the process is complete, you will be prompted with the message: **The data is ready for export or analysis. Would you like me to download it as a CSV?.**

Type **Yes** to proceed with downloading the dataset.



Click on **Download** to save the data to your local machine.

5. Analyze the report

This tool allows you to view details about the model used to generate the synthetic data.

Copy and paste the below prompt to see the report

plaintext

Provide a detailed report on the model used for generating the synthetic data. Include information about the modeling approach, data distribution handling, and feature relationships learned. Also include performance metrics such as accuracy, AUC (Area Under the Curve), and any validation metrics used to assess the quality and similarity of the synthetic data relative to the original dataset. Present the report in a clear and structured format suitable for analysis.

After you give the prompt, you will see a detailed report along with visualizations.

The screenshot displays the Tonic Fabricate web application. On the left is a dark sidebar with navigation options: 'Default', 'insurance', 'PLANS', 'WORKFLOWS', and 'API'. The main area shows a table titled 'insurance' with 1338 records. The table has columns: id, age, gender, bmi, children, smoker, region, and expenses. Below the table are navigation controls: '- Previous', 'Page 1 of 14', and 'Next ->'. On the right, a 'Synthetic Insurance Data Generator' panel is visible. It contains a small table with age groups and costs, a text box explaining a monotonic increase in expenses with age, a 'Summary Scorecard' table, and a concluding statement about the dataset's similarity to the original. At the bottom of this panel is a text input field with the placeholder 'What would you like to do next?' and a 'Agent' dropdown.

id	age	gender	bmi	children	smoker	region	expenses
1	33	male	29.2	3	no	southeast	5275.43
2	42	male	32.9	3	no	southeast	16588.7
3	43	female	38.6	2	no	southwest	4876.97
4	38	female	24.7	1	no	southwest	2868.68
5	62	female	28.3	1	no	northeast	14427.8
6	52	male	34	0	no	southwest	2065.15
7	63	female	39.9	0	no	southwest	23015.9
8	44	male	35.6	1	no	southwest	15470.2
9	53	male	35.7	0	yes	southeast	34589.7
10	24	male	26.2	3	no	northeast	4704.95
11	56	female	33	0	no	southeast	16720.5
12	24	female	27.8	1	no	southeast	1970.8
13	40	male	31.5	0	no	northeast	7378.22
14	47	male	33	0	yes	southwest	17645.3
15	49	male	31.2	1	no	southwest	11600.3

Evaluation Dimension	Weight	Score
Numeric Distribution (KS Tests)	30%	96.8%
Categorical Distribution (TVD)	20%	99.0%
Correlation Preservation	25%	96.3%
Discriminator AUC	25%	98.4%
Overall Similarity Score	100%	97.5%

The synthetic dataset exceeds the 80% similarity target across all dimensions, scoring 97.5% overall. The data is statistically consistent with the source, preserves all key relationships, and is indistinguishable from the original by a machine learning classifier.

Conclusion

Congratulations! You have completed the lab on data augmentation using iLoveCSV and Tonic Fabricate tools.